Homework 3 — changed numbers (extra practice)

Module 3 · Homework · No date

The numbers on this sheet are changed from her sheet.

Name _____ Date _____

An antibiotic with a clearance of 4.62 L/hr and an apparent volume of distribution of approximately 30 L is to be administered by IV infusion to attain a steady-state plasma concentration of 15 mcg/mL.

a Module 3

What is an appropriate rate of infusion to achieve the desired steady-state concentration of 15 mcg/mL?

b Module 3

What is the half-life of the drug that is to be infused?

c Module 3

What is the expected concentration of drug in the plasma 6 hours after the start of the infusion?

d Module 3

How much time is required to reach 95% of the steady-state concentration if no loading dose were used?

e Module 3

Recommend a loading dose to obtain the steady-state concentration immediately.

f Module 3

If the infusion were stopped after the desired steady-state concentration of 15 mcg/mL was attained, what would the concentration be 8 hours after the cessation?

g Module 3

If the infusion rate that you recommended in a (above) were doubled, what would the new steady-state concentration be?

h Module 3

If the infusion rate that you recommended in a (above) were doubled, how long would it take to reach 95% of the new steady-state concentration (no loading dose)?

i Module 3

If the infusion were stopped after proceeding long enough at the rate determined in part g (above) to reach the new steady-state concentration, what would the concentration be 6 hours after the cessation?

j Module 3

What specific change would you make to achieve a steady state concentration of 9 mg/L? Give the new infusion rate.